LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 4 Test (Set B): Carbon and its Compounds

Chapter / Topic: Carbon and its Compounds

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Write balanced chemical equations and structural formulae wherever asked.

SECTION A

6 marks

Choose the correct option.

Q1. The next homologue of methane (CH_4) is: **[1]**

(a) C_2H_4 (b) C_3H_8
(c) C_2H_6 (d) CH_3OH

Q2. The functional group present in propanal is: **[1]**

(a) alcohol (b) ketone
(c) carboxylic acid (d) aldehyde

Q3. The two carbon atoms in ethene are joined by: **[1]**

(a) a single bond (b) a double bond
(c) a triple bond (d) an ionic bond

Q4. Soaps are: **[1]**

(a) esters of ethanol
(b) ammonium salts of short-chain acids
(c) sodium or potassium salts of long-chain carboxylic acids
(d) alcohols

Q5. Vegetable ghee is made from vegetable oils by: **[1]**

(a) hydrogenation (b) oxidation
(c) substitution (d) esterification

Q6. **Assertion (A):** Pure ethanoic acid is called glacial acetic acid.
Reason (R): Pure ethanoic acid freezes in cold climates because its melting point is 290 K. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Why do covalent compounds generally have low melting and boiling points? Why are they poor conductors of electricity? **[2]**

OR

Name these compounds: (a) $CH_3–CH_2–Br$, (b) HCHO, (c) CH_3COCH_3, (d) $CH_3CH_2CH_2OH$.

Q8. What is a homologous series? Give two of its characteristics. **[2]**

SECTION C

6 marks

Q9. Answer the following: **[3]**

(i) Write the general formula of alkenes. **[1]**

(ii) Write the name and formula of the third member of the alkane series. **[1]**

(iii) By how much do the molecular masses of two successive members of a homologous series differ? **[1]**

Q10. What is a substitution reaction? Give a balanced equation and the condition needed. **[3]**

OR

Why do soap molecules form micelles in water? Would micelles form in ethanol? Explain.

SECTION D

5 marks

Q11. Answer the following about ethanoic acid: **[5]**

(i) Write balanced equations for its reactions with (a) sodium hydroxide, (b) sodium carbonate, (c) ethanol (in the presence of an acid). **[3]**

(ii) Give one use of ethanoic acid and one use of esters. **[2]**

OR

(a) What are catenation and tetravalency? [2]
(b) Write the structural formulae of ethane, ethene and ethyne. [3]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Hard water contains salts of calcium and magnesium. When soap is used in hard water, it reacts with these salts to form an insoluble substance called scum, and it does not lather well. Detergents are generally ammonium or sulphonate salts of long-chain carboxylic acids. Their charged ends do not form insoluble precipitates with calcium and magnesium ions, so they work well even in hard water. **[4]**

(i) Which ions make water hard? **[1]**

(ii) What is scum? **[1]**

(iii) Why are detergents better than soaps for washing in hard water? Give two uses of detergents. **[2]**

~ All the best! ~